

Testing Strategy

Current status

Before Gateway/platform hardening, the solution had no test project and no Moq usage. It now contains tests/FinWallet.Application.Tests using:

- xUnit v3;
- Moq;
- .NET 8 / Microsoft Testing Platform.

The first strict-mock test targets OpenBankAccountHandler, mocks IWalletStore, IBankAccountStore and IBankProvider, and proves that the external bank provider is not called when the owned wallet is missing.

What should be mocked?

Application orchestration tests may mock boundaries whose behavior is not the subject of the test:

- IBankProvider;
- IExternalFraudProvider;
- ICommunicationGateway;
- persistence interfaces;
- TimeProvider for time-dependent behavior.

Strict mocks are preferred for financial orchestration because an unexpected provider/database call may be a meaningful defect.

What cannot be proven with mocks?

The following require real infrastructure integration/concurrency tests:

- MSSQL rollback;
- Serializable / UPDLOCK / HOLDLOCK;
- unique-key races;
- refresh-token compare-and-set;
- deterministic wallet lock ordering;
- idempotency range locking;
- persisted Debit=Credit aggregate validation;
- connection-pooling behavior;
- Redis Lua atomicity/reconnect/fail-closed behavior;
- YARP routing/auth/health/load balancing;
- Gateway direct-bypass rejection.

Required integration suite

At minimum:

- 1 concurrent duplicate registration;
- 2 concurrent refresh rotation;
- 3 ten transfers with the same Idempotency-Key;

- 4 same key + different payload;
- 5 one hundred concurrent debits from one source wallet;
- 6 opposite-direction A->B and B->A deadlock test;
- 7 Redis unavailable during OTP verify;
- 8 FakeFraud timeout before transfer posting;
- 9 Gateway request without JWT;
- 10 backend request without downstream key;
- 11 rate-limit rejection;
- 12 ledger imbalance injection -> rollback.

Financial invariant test

Load tests must not trust HTTP success counts alone:

```
Initial balances
+ durable credits
- durable debits
= final balances
```

```
SUM(Ledger Debit) = SUM(Ledger Credit)
```

Successful FinancialTransactions, IdempotencyRecords and LedgerJournals must also reconcile.

CI policy

CI runs:

```
restore
-> Release build --warnaserror
-> dotnet test --no-build
```

A new project is not considered complete until it is included in FinWallet.sln.

Current coverage assessment

Unit-test infrastructure now exists, but coverage is still at an early stage. “Moq is present” does not mean real MSSQL/Redis/YARP correctness has been proven.